

FMUSER FUQ-44105B QAM Modulator 16-24 DVB-S2 Tuner 128 IP 1 ASI Input 16 RF 16 IP Output DVB-S2 IP ASI QAM Multiplexing Scrambling EN300 429 ITU-T J.83A-B For CATV Broadcasting

Product overview



The FMUSER FUQ-44105B is a QAM modulator with integrated multiplexing and scrambling. It accepts inputs from up to 24 DVB-S2 tuners, 128 IP sources via GE1, and ASI, and outputs 16 RF carriers and 16 IP (MPTS) streams via GE2. The device processes signals through multiplexing, scrambling to ETR289/ETSI standards, and QAM modulation. Managed via web interface, it is ideal for modern CATV broadcasting systems. FU-44105C is also characterized with high integrated level, high performance and low cost. It supports dual power supply (optional). This is very adaptable to newly generation CATV broadcasting system.

Key feature

Supports up to 24 DVB-S2 tuners, 128 IP inputs over UDP/RTP, and ASI, with TS file playback via USB.

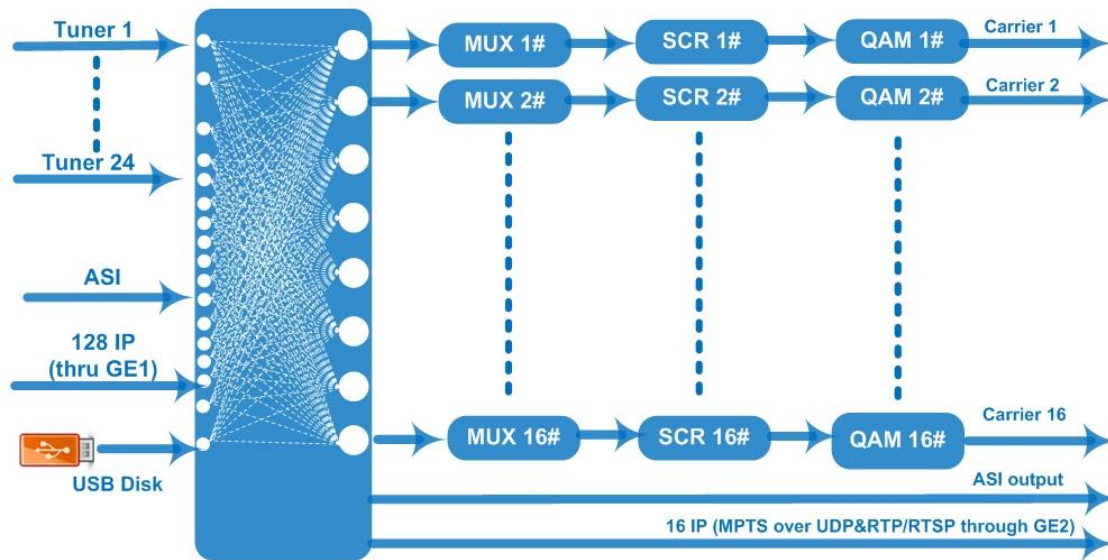
Delivers 16 non-adjacent RF outputs and 16 IP (MPTS) outputs for versatile distribution.

Integrates multiplexing, scrambling to ETR289/ETSI standards, and QAM modulation with MER $\geq$ 40dB.

Enables PSI/SI editing, PCR adjusting, and web management for configuration.

Offers optional redundant power supply for enhanced reliability.

Working Principle



Specifications

Input	24 or 16 DVB -S/S2/S2X FT A Tuner input as per order	24 or 16 DVB -S/S2/S2X FT A Tuner input as per order	24 or 16 DVB -S/S2/S2X FT A Tuner input as per order	24 or 16 DVB -S/S2/S2X FT A Tuner input as per order	24 or 16 DVB -S/S2/S2X FT A Tuner input as per order
Input	128 IP input through GE1 over UDP and RTP protocol	128 IP input through GE1 over UDP and RTP protocol	128 IP input through GE1 over UDP and RTP protocol	128 IP input through GE1 over UDP and RTP protocol	128 IP input through GE1 over UDP and RTP protocol
Input	1 ASI input, BNC interface	1 ASI input, BNC interface	1 ASI input, BNC interface	1 ASI input, BNC interface	1 ASI input, BNC interface
Input	TS files input and playing via the USB disk (FAT 32)	TS files input and playing via the USB disk (FAT 32)	TS files input and playing via the USB disk (FAT 32)	TS files input and playing via the USB disk (FAT 32)	TS files input and playing via the USB disk (FAT 32)
Input	Tuner	DVB-S	Frequency In	Frequency In	950~2150MHz
Input	Tuner	DVB-S	Symbol rate	Symbol rate	0.5~45Msps
Input	Tuner	DVB-S	Signal Strength	Signal Strength	- 65~-25dBm
Input	Tuner	DVB-S	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8
Input	Tuner	DVB-S	Constellation	Constellation	QPSK
Input	Tuner	DVB-S	Max input bitrate	Max input bitrate	≤129 Mbps

Input	Tuner	DVB-S2	Frequency In	Frequency In	950~2150MHz
Input	Tuner	DVB-S2	Symbol rate	Symbol rate	QPSK/8PSK / 16APSK: 0.5~45 Msps 32APSK: 0.5~40Msps;
Input	Tuner	DVB-S2	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10
Input	Tuner	DVB-S2	Constellation	Constellation	QPSK, 8PSK, 16APSK, 32APSK
Input	Tuner	DVB-S2	Max input bitrate	Max input bitrate	≤129 Mbps
Input	Tuner	DVB-S2X	Frequency In	Frequency In	950-2150MHz
Input	Tuner	DVB-S2X	Symbol rate	Symbol rate	QPSK/8PSK /16APSK: 0.5~45 Msps 8APSK/32APSK: 0.5~40Msps

Input	Tuner	DVB-S2X	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10, 2/3-L, 32/45, 11/15, 7/9
Input	Tuner	DVB-S2X	Constellation	Constellation	QPSK, 8PSK, 8APSK, 16APSK, 32APSK
Input	Tuner	DVB-S2X	Max input bitrate	Max input bitrate	≤129 Mbps
Multiplexing	Max PID Remapping	Max PID Remapping	360 output per channel	360 output per channel	360 output per channel
Multiplexing	Function	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Function	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically

Scrambling Parameters	Max simulcrypt CA	Max simulcrypt CA	4	4	4
Scrambling Parameters	Scramble Standard	Scramble Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling Parameters	Connection	Connection	Local/remote connection	Local/remote connection	Local/remote connection
Modulation	QAM Channel	QAM Channel	16 non-adjacent carriers output	16 non-adjacent carriers output	16 non-adjacent carriers output
Modulation	Standard	Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B
Modulation	MER	MER	≥40db	≥40db	≥40db
Modulation	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	RF output level	RF output level	-20~+10dbm(87~107 dbμV), 0.1db step	-20~+10dbm(87~107 dbμV), 0.1db step	-20~+10dbm(87~107 dbμV), 0.1db step
Modulation	Symbol Rate	Symbol Rate	5.0Msps~7.0 Msps, 1ksps stepping	5.0Msps~7.0 Msps, 1ksps stepping	5.0Msps~7.0 Msps, 1ksps stepping
Modulation			J.83A	J.83A	J.83B
Modulation	Constellation	Constellation	16/32/64/128/ 256QAM	16/32/64/128/ 256QAM	64/256 QAM
Modulation	Bandwidth	Bandwidth	8M	8M	6M
Stream out	16 RF output (F type interface)	16 RF output (F type interface)	16 RF output (F type interface)	16 RF output (F type interface)	16 RF output (F type interface)

Stream out	16 IP (MPTS) output over UDP and RTP/RTSP (thru GE2), 2 ASI output(one as mirror)	16 IP (MPTS) output over UDP and RTP/RTSP (thru GE2), 2 ASI output(one as mirror)	16 IP (MPTS) output over UDP and RTP/RTSP (thru GE2), 2 ASI output(one as mirror)	16 IP (MPTS) output over UDP and RTP/RTSP (thru GE2), 2 ASI output(one as mirror)	16 IP (MPTS) output over UDP and RTP/RTSP (thru GE2), 2 ASI output(one as mirror)
System	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)
System	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language
System	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
General	Dimension (W *D*H)	Dimension (W *D*H)	Dimension (W *D*H)	482mm×230 mm×44mm	482mm×230 mm×44mm
General	Temperature	Temperature	Temperature	0~45■(Operation) ; -20~80 ■(Storage)	0~45■(Operation) ; -20~80 ■(Storage)
General	Power	Power	Power	AC 100V±105 0/60Hz; AC 220V±10%, 50/60HZ	AC 100V±105 0/60Hz; AC 220V±10%, 50/60HZ

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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